Assignment 1: Network Latencies, Ping & Traceroute

CS 422: Computer Networks, Fall 2026
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Due Date: Tuesday, September 08, 2026 @ 23:45 Eastern Time

1. Ping Test and Round-Trip Time (RTT) (50 points)

(a) For each ip address listed here: https://iperf3serverlist.net/, including your own ip address, do the following:

  (i) Run a ping test and find the min, max, and average round-trip time (rtt).

  (ii) Find the geolocation coordinates from publicly available data.

(b) Do a scatter plot: distance vs rtt, where distance corresponds to the geographical distance between your location and the destination ip address location. Each data point corresponds to a destination ip address.

(c) Explain whether and how distance relates to rtt. What do min and max rtt convey about the distance, and the network state?

2. Latency Breakdown (50 points)

(a) Pick 5 random ip addresses from the list of ip address here: https://iperf3serverlist.net/. Find the round-trip time from your machine to each intermediate hop along the path towards destination ip addresses using traceroute. Your script could filter out non-responsive hops.

(b) Plot a stacked bar chart showing the breakdown of latencies to each hop, corresponding to each of the five chosen destination ip addresses.

(c) Do a scatter plot: hop count vs rtt. Each data point corresponds to a destination ip address.

(d) Explain your observations from these plots. How does hop count relate to rtt?

Note: It is fine to skip servers that are non-responsive (your script should handle it). You may also see "*" in the output of traceroute, which indicates that the hop is not responding to icmp. This is fine, as long as the traceroute completes with the final destination responding to icmp. In other cases, traceroute will timeout, and the script should mark it as non-responsive.

Report

Please include a github/gitlab link in your report: This assignment is mainly related to coding, so include a link or line ranges to the relevant code section in your repo (e.g., a specific function or class). For plots and visualizations, please include the plots in the report and also point to the relevant code sections that generate the plots.
These experiments should be fully automated using a script (e.g., Python, Bash, etc.) that can also handle corner cases such as non-responsive servers. The input to the script should be a file containing the list of ip addresses to ping and traceroute, any other command line arguments if needed. The script should also invoke plotting, and generate pdf plots all at once. Please include a github/gitlab link in your report. The codebase should include all inputs and scripts required to re-run the experiments and generate plots (all in one-shot).
Note: Grading is still based on the oral exam during PSO/office hours in the week of *September 14*.
Grace period: 3 days (72 Hours) grace period is allowed for submission. After the grace period, 25% off for every 24 hours late, rounded up.